Interactive Dashboard & Tableau Storytelling Assignment

Q1. Difference Between a Tableau Dashboard and a Tableau Story Point

Tableau Dashboard	Tableau Story Point
A dashboard is a collection of multiple worksheets, charts, and visualizations displayed on a single screen.	A story point is a sequence of worksheets and dashboards used to present information step-by-step.
Used for interactive monitoring and analysis.	Used for storytelling and presenting insights in a logical flow.
Focuses on real-time exploration of data.	Focuses on explaining findings, trends, and conclusions.
Users can interact with filters, parameters, and actions.	Users move through story points to understand a narrative.

Healthcare Use Cases

Dashboard Use Cases

1. **Hospital Performance Dashboard**
    - Displays patient admissions, bed occupancy, emergency room visits, and average treatment time in one view.
    - Helps administrators monitor hospital operations.
2. **Patient Monitoring Dashboard**
    - Shows patient demographics, disease distribution, treatment outcomes, and readmission rates.
    - Assists doctors and management in tracking healthcare quality.

Story Point Use Cases

1. **Disease Outbreak Analysis**
    - Story Point 1: Number of reported cases.
    - Story Point 2: Geographic spread of the disease.
    - Story Point 3: Impact on different age groups.
    - Story Point 4: Recommended interventions.
2. **Patient Readmission Study**
    - Story Point 1: Overall readmission rate.
    - Story Point 2: Departments with highest readmissions.
    - Story Point 3: Factors causing readmissions.

Q2. What are Actions in Tableau?

Actions are interactive features in Tableau that allow users to perform specific operations by clicking, hovering, or selecting marks in a visualization. They enable dashboards to become dynamic and responsive.

Common Types of Actions

1. **Filter Action** – Filters data in one view based on selections made in another view.
2. **Highlight Action** – Highlights related data points across views.
3. **URL Action** – Opens a web page when a mark is selected.
4. **Parameter Action** – Changes parameter values through user interaction.
5. **Set Action** – Updates set membership dynamically.

How Filter Action Helps Hospital Management Analyze Patient Data

A **Filter Action** allows hospital managers to select a specific category and automatically filter related information across other charts.

Example Scenario

Suppose a hospital dashboard contains:

- Patient Admissions by Department
- Average Length of Stay
- Patient Demographics
- Treatment Success Rate

When a manager clicks on the **Cardiology** department in the admissions chart:

- All other charts automatically update to show only Cardiology patients.
- Management can analyze:
    - Number of admitted patients
    - Average stay duration
    - Age and gender distribution
    - Recovery and readmission rates

Benefits

- Faster identification of trends and problem areas.
- Improved resource allocation across departments.
- Better monitoring of patient outcomes.
- Enhanced decision-making without manually applying filters.

Example: If Cardiology shows a high readmission rate after filtering, hospital administrators can investigate treatment processes and allocate additional resources to improve patient care.

Q10 – Insight-Based Analysis from Dashboard Visualizations

1. Which department generates the highest treatment cost?

Neurology Department

- Neurology has the highest total treatment cost:
  ≈ 5,320,474
- It is higher than all other departments:
  - Cardiology (~4.4M)
  - Pediatrics (~3.7M)
  - Orthopedics (~3.1M)
  - General Medicine (~3.2M)

✓ **Insight:** Neurology is the most resource-intensive department.

2. Which region has more critical patients?

West Region

- West region has the highest patient count (**61 patients** overall).
- Based on distribution and volume, it is the most likely region contributing the highest number of critical cases.

✓ **Insight:** West region carries the highest patient load and critical care demand.

3. What business decisions can hospital management take from these insights?

1. Optimize high-cost department (Neurology)

- Review treatment expenses
- Improve efficiency in resource usage
- Evaluate cost vs outcome performance

2. Strengthen West region infrastructure

- Increase hospital staff and specialists
- Expand ICU and emergency facilities
- Improve critical care response capacity

3. Improve patient outcome management

- 32% patients are in critical condition
- Introduce:

    - Early diagnosis systems
    - Faster treatment protocols
    - Better monitoring systems

4. Better resource allocation

- Deploy more doctors to:
    - Neurology (high cost)
    - West region (high patient load)

5. Cost control strategy

- Identify cost-heavy procedures
- Ensure high spending is linked to better outcomes

Final Insight Summary

- Highest cost department: **Neurology**
- Region with highest critical demand: **West**
- Key action: **Optimize Neurology + strengthen West region healthcare capacity**